

CSCC01 Summer 2026
Group Project Proposal: CRMP
Group Name: Five Guys

Personas

Personas 1: Public User

Name: Mark Thompson

Age: 62

Occupation: Retired investment banker in a rural community

Status: Married

Location: Muskoka, Ontario

Personality: He values tradition; likes chatting in person with others; looks serious but is kind, outspoken, and community oriented; persistent when things affect him.

Education and Experience: He has a bachelor's degree in economics from Queens university. He has had decades of experience working in investment banking in big firms in downtown Toronto even taking on senior advisory roles. Now as a retiree he thinks more in a practical sense based on local experiences than a financial sense.

BIO: Mark is a retired investment banker who moved closer to nature after his retirement. After a long time in finance, he now enjoys a slower paced life. Mark is very community-oriented and loves chatting with his neighbors and others. He does value tradition and doesn't fully trust decisions from the government that affect his lifestyle. He seems serious at first, but is kind, and approachable when you get to know him. However, he is very persistent when something affects him directly and will try to solve the issue.

GOALS: Mark wants to avoid having to travel so far for government related activities because they are in such a big riding. Ensure rural communities have a fair voice in electoral and civic decisions

FRUSTRATIONS: Currently Mark and his neighbors have to travel a very long way to get to the nearest voting area. They also feel like that since they are in a large riding they don't have much say as they are in a more rural part of the community. He also finds traditional documents of ridings and ways to send complaints very tedious and unnecessarily hard.

INTEREST IN PRODUCT: Mark would like to use a convenient tool to voice his complaint about ridings to the government.

0 - 5 scale:

- Incentive - 4
- Fear - 3
- Achievement - 4
- Growth - 3

- **Power - 4**
- **Social - 4**

QUOTES: “People making these maps should understand what it’s actually like to live out here.”,

“Why do these guys (the government) keep making things more complicated?”

DOMAIN AWARENESS: Not much. He occasionally uses his phone and a home computer, but mostly uses pen and paper.

TECH KNOWLEDGE: Has very little experience about computers and the web from his later years before retirement. But he has time to learn.

Personas 2: Commissioner

Name: Richard Jefferson

Age: 50

Occupation: Commissioner

Status: Married

Location: Ottawa, Ontario

Personality: analytical; follows the book; procedural; calm under pressure; loves structured processes;

Education and Experience: He has a bachelors in geography from McGill and a masters in public administration from the university of Toronto. He used to work in many federal roles involving urban planning, data analysis and more before becoming a commissioner. Richard is basically an expert at data, and in parallel is very good with technology.

BIO: Richard is a boundaries commissioner who is responsible for reviewing the public input for the redistribution processes. He is a very analytical, detail oriented, and procedural commissioner. He goes by the book and values fairness and integrity. He makes sure the proposed map follows all requirements while factoring in public opinion.

GOALS: Richard wants a convenient and structured way to do his job within the redistribution process. He wants an easier way to view all public input for the proposed boundary map and be able to compare it with the proposed map. He also wants to be able to easily view data required for the proposed map or counter proposals. One such piece of data could be the population distribution. He would also benefit from an easy one stop shop for all things related to the redistribution plan.

FRUSTRATIONS: Currently public submissions are not centralized and are coming from email, mail and many other places. Some public submissions also lack proper specifications or data required in the submission. It is also annoying to compare submissions to the actual proposed map, while also needing to do checks such as recalculating data to see if the public submission even makes sense at all. He also needs actual public submissions from ones that aren't serious. All the data within the entire redistribution process is also not centralized and is in many places making it hard to compile all the data together and export or send it over.

INTEREST IN PRODUCT: Richard is an example of the commissioner side of the product. Richard would be interested in a product that can make his job easier. He would have centralized public submissions that are easy to filter, tag, and comment on. The product would also make data compilation easier for Richard.

0 - 5 scale:

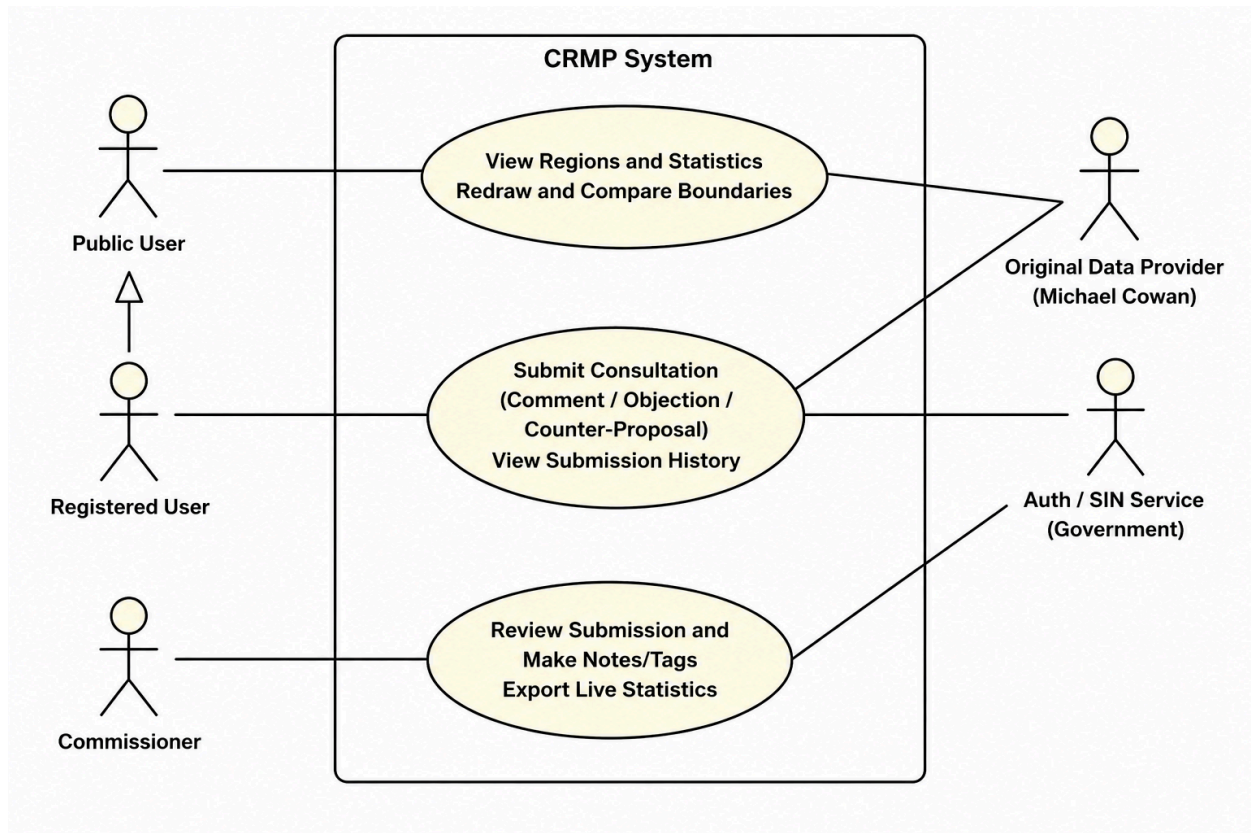
- Incentive - 5
- Fear - 1
- Achievement - 5
- Growth - 3
- Power - 3
- Social - 3

QUOTES: "If we can't trace a submission back to a map feature, it isn't valid",
"Public submissions only work if we can efficiently interpret and understand what people"

DOMAIN AWARENESS: Richard understands everything needed for the redistribution from the data to the laws.

TECH KNOWLEDGE: Expert in data analysis, and in parallel is very good with software and computers.

Use Case Diagram



View Regions and Statistics, Redraw and Compare Boundaries

This use case allows a public visitor to open the map, browse proposed electoral-district boundaries, and click on a riding to view read-only statistics. It is accessible to both anonymous visitors and registered users. Users can toggle the existing legally in force boundaries for comparison and temporarily update or redraw boundaries without making a stored submission. To function, this activity relies on the Original Data Provider to supply the underlying map structures and contextual statistical data.

Submit Consultation (Comment / Objection / Counter-Proposal), View Submission History

This use case enables a registered public user to submit a written comment, a boundary objection, or a counter-proposal. Before a submission can continue, the system checks the user's registration status and validates identity information, such as a SIN, through an external Auth/SIN Service. Upon successful validation, the system stores the submission, timestamps it, assigns a unique public reference number, and makes the current status visible on the user's "My Submissions" page. The structural data necessary for formulating and calculating counter-proposals is drawn from the Original Data Provider.

Review Submission and Make Notes/Tags, Export Live Statistics

An authenticated commissioner logs in to an administrative dashboard to review public submissions and collaborate on region-specific work. Within this workspace, commissioners can apply tags and internal notes that remain visible only to other commissioners, while inspecting map changes and live consultation statistics. The system allows commissioners to export a filtered subset of data in the final report.

Use Case Scenarios

Use Case 1: Public User View and Redraw Activity

1. Description of the use case:

A public visitor can open the map (CRMP) as a public user, browse proposed electoral-district boundaries, toggle the existing legally in force boundaries for comparison, and click on a riding to view read-only statistics. This activity should be available to both anonymous users and registered users. Submissions cannot be created in this use case, and the map panel should remain both informational and interactive.

2. Business Event Name:

Public User Views Proposed Boundaries and Redraw/Reselect any regions/communities based on the data structure provided with the UI.

3. Business Use Case Name and Number:

Public User View and Redraw Activity (Use Case 1)

4. Trigger:

A public user opens the interactive map and/or selects a riding or boundary segment to inspect.

5. Preconditions:

The public-facing map and proposed boundary data are available. The user may be anonymous or registered. No login is required for viewing.

6. Postcondition:

No system data is created, modified, or deleted. The user has viewed the proposed boundaries and related read-only information.

7. Interested Stakeholders:

Public users, commissioners.

8. Active Stakeholders:

Public user, including anonymous visitor or registered user.

9. Normal Case Steps:

1. The public user opens the public-facing map interface.
2. The system displays the 'proposed electoral-district boundaries' (based on GPS, postal codes, etc. or just the city center) as the default screen.
3. The user can click a riding on the map for boundary editing and statistics viewing.
4. The system also opens a read-only side panel showing the riding's statistics, including population and other required contextual data.

5. The user can update (redraw) the displayed boundaries and statistics without making any submission or stored change.
6. The user optionally toggles (switches to) the existing map or the edited map for comparison.

10. Alternatives:

Alternative 1: Anonymous Browsing

A1.1: The user opens the map without signing in.

A1.2: The system allows browsing and comparison of map and riding statistics. -> Go to Step 4

A1.3: The user remains anonymous throughout this use case. -> Go to Step 3

Alternative 2: Registered User Browsing

A2.1: The user opens the map while already signed in.

A2.2: The system shows the same read-only map and statistics. -> Go to Step 4

A2.3: The user continues browsing without creating or modifying any submission. -> Go to Step 3

Alternative 3: Map Comparison (New Page May Need for Redrawing)

A3.1: The user can toggle between proposed boundaries and existing legal boundaries any time. -> Go to Step 6

A3.2: The system updates the visible map (region boundaries) each time.

A3.3: The edited map can be shown in the same page or another page depending on the screen size.

11. Exceptions:

Exception 1: Map or Statistics Fail to Load

E1.1: The user opens the map, but the map layer or riding statistics cannot be retrieved.

E1.2: The system displays an error and stops the viewing session.

E1.3: No data is changed. -> End Use Case

Use Case 2: Public User Submit Activity

1. Description of the use case:

A registered public user submits a public consultation activity, which may be a written comment, a boundary objection, or a counter-proposal. The user must be logged in, and the submission is associated with the user's account, timestamped, assigned a unique public reference number, and shown on the user's "My Submissions" page.

2. Business Event Name:

Public User Submits a Consultation, including a comment, and/or an objection, and/or a counter-proposal

3. Business Use Case Name and Number:

Public User Submit Activity (Use Case 2)

4. Trigger:

A registered public user chooses to submit a comment, objection, or counter-proposal from the public-facing interface.

5. Preconditions:

The public consultation window is open. The user is logged in with a verified registered account. A legitimate mechanism exists for verifying the user's SIN, and the user has either provided the SIN or may add it as optional registration information before or during later account use.

6. Postcondition:

A new submission is stored, timestamped, assigned a unique public reference number, associated with the submitter's account, and visible on the user's "My Submissions" page with its current status.

7. Interested Stakeholders:

Public users, commissioners.

8. Active Stakeholders:

Registered public user.

9. Normal Case Steps:

1. The registered public user opens the public map and chooses any region to submit feedback.
2. The user can select the relevant riding or boundary segment, and/or draw the counter-proposal geometry when needed, as similar to Step 5 in Use Case 1.
3. The user enters the submission content and the system checks the user's registration status before submission can continue.
4. The user provides or confirms the identity-recognition information, including SIN if applicable.

5. The system validates the account, the required fields, and the submission rules for the selected submission type, once the user presses the submit button.
6. The system stores the submission, timestamps it, assigns a unique public reference number, and associates it with the user's account.
7. The system makes the submission visible on the user's "My Submissions" page with its current status.

10. Alternatives:

Alternative 1: User Submits a Written Comment or Boundary Objection

A1.1: The user selects a riding or boundary segment.

A1.2: The user enters free text or a structured objection. -> Go to Step 2

A1.3: The system records the submission as a comment or objection and shows it on "My Submissions." -> Go to Step 7

Alternative 2: User Submits a Counter-Proposal

A2.1: The user draws an alternative boundary configuration by reassigning geographic units. -> Go to Step 2

A2.2: The system calculates the effect of the changes on population balance in real time.

A2.3: The user continues refining the proposal until the result is acceptable (checked by front-end), then submits it.

Alternative 3: User Completes Registration First

A3.1: The user is not yet authenticated when starting from the public map.

A3.2: The system routes the user through login or registration before submission can continue.

A3.3: After successful login, the user resumes at Step 3. -> Go to Step 3

11. Exceptions:

Exception 1: Missing or Invalid SIN, or Mass Similar Submissions

E1.1: The user attempts to submit without a SIN, with an invalid SIN, or while sending a large set of highly similar reports.

E1.2: The system blocks the submission.

E1.3: No submission is created. -> End Use Case

Exception 2: Counter-Proposal Violates Structural Rules

E2.1: The user attempts to submit a counter-proposal that is not contiguous or otherwise structurally invalid.

E2.2: The system blocks the submission before Step 2. -> End Use Case

Use Case 3: Commissioner Collaborative Work Activity

1. Description of the use case:

A commissioner logs in to a backend dashboard to review submissions, apply tags and internal notes visible only to other commissioners, inspect map changes and live consultation statistics, and collaborate on region-specific work without entering any extra identification information. The collaboration flow also supports viewing map changes similarly to the public interface, applying and undoing one or more changes at any time, and comparing changes of the data aggregation behavior at different map granularities.

2. Business Event Name:

Commissioner Collaborates on Regional (e.g. Provincial) Review

3. Business Use Case Name and Number:

Commissioner Collaborative Work Activity (Use Case 3)

4. Trigger:

A commissioner signs in to the administrative dashboard with verified role (and/or assigned region) and begins reviewing or collaborating on region-specific submissions and map work.

5. Preconditions:

The user is an authenticated commissioner with dashboard access. The region's collaboration space and public submissions for that region are available. No extra commissioner identity entry is required beyond the authenticated account.

6. Postcondition:

The commissioner's tags, notes, exports, and map-change decisions are stored or reflected in the system, and the dashboard statistics (shared workspace) remain updated for the region.

7. Interested Stakeholders:

Commissioners, public users, the commission's final reporting process.

8. Active Stakeholders:

Commissioner.

9. Normal Case Steps:

1. The commissioner logs in to the administrative dashboard.
2. The system displays the real-time consultation overview for the commissioner's region.
3. The commissioner opens a riding and/or specific submission queue (depending on the design or the system) and reviews the submissions in context.
4. The commissioner views the map and compares current boundaries with proposed changes while inspecting the associated regional statistics.
5. The commissioner adds tags or internal notes, which remain visible only to other commissioners at the desired granularity.

6. The system updates the region's data aggregation behavior and live statistics as changes are applied.
7. The commissioner can undo one or more changes if needed and continues collaborating on the region.
8. The commissioner saves the work, leaving the updated state available for further commissioner collaboration.

10. Alternatives:

Alternative 1: Drill Into a Riding and Process Sequentially

A1.1: The commissioner selects a riding from the dashboard.

A1.2: The system shows the full queue of submissions for that riding.

A1.3: The commissioner processes the queue sequentially with tagging support. -> Go to Step 5

Alternative 2: Export a Filtered Subset for the Final Report

A2.1: The commissioner filters submissions by riding, date, or type.

A2.2: The commissioner chooses CSV or PDF export.

A2.3: The system exports the filtered subset with associated tags for inclusion in the final report.

-> Go to Step 8

Alternative 3: Compare Regions with Different Granularities

A3.1: The commissioner changes the granularity of the region edited or compared.

A3.2: The system shows how the data aggregation behavior changes at that granularity.

A3.3: The commissioner continues analysis or reverts the change. -> Go to Step 7

Alternative 4: Concurrent Multi-Commissioner Collaboration

A4.1: Two or more commissioners open the same regional workspace and begin reviewing the same riding, map area, or submission set concurrently.

A4.2: Each commissioner views their own statistics, comparison panels, and analytical views independently.

A4.3: A commissioner creates, modifies, or removes an internal tag, note, annotation, or review status.

A4.4: The system propagates the update to all commissioners currently viewing the same regional workspace.

A4.5: If multiple commissioners modify the same tag, note, annotation, or review item simultaneously, the system detects the conflict and displays a conflict notification showing the competing versions.

A4.6: The commissioners select a version, merge the changes, or create a new revision.

A4.7: The system records the resolved version and updates all active collaboration sessions.

A4.8: Commissioners continue collaborating on submissions, map changes, and consultation data -> Go to Step 5.

11. Exceptions:

Exception 1: Unauthorized Access

E1.1: A user without commissioner privileges attempts to open the collaborative dashboard.

E1.2: The system denies access. -> End Use Case

Exception 2: Collaboration Conflict Cannot Be Resolved Automatically

E2.1: Two or more commissioners submit incompatible updates to the same note, tag, or annotation.

E2.2: The system cannot automatically merge the changes.

E2.3: The system preserves all conflicting versions and marks the item as requiring manual resolution.

E2.4: The conflicting item remains visible to collaborating commissioners and blocks for saving and exiting until a resolution is selected. -> End Use Case.

UI sketches

Please zoom in to see the clear sketches!

background

Proposed boundary map

The proposed boundary is a dashed line. The map shows the Yukon and Northwest Territories. A legend box in the bottom left corner contains the following information:

- Proposed boundary
- Yukon
- Northwest Territories
- Other provinces and territories

Current boundary map

The current boundary is a solid line. The map shows the Yukon and Northwest Territories. A legend box in the bottom left corner contains the following information:

- Current boundary
- Yukon
- Northwest Territories
- Other provinces and territories

toggle the map
on this page

Account #: _____
Password: _____
forget your password?
Don't have an account? *Sign up*

Back

Enter your registered email:

Next

The user receives a verification code / link to update password

Sign Up

First name : _____

Last name : _____

Province : _____ ▼ ← select list

Account # : _____

Password : _____

Confirm password : _____

Phone number : fixed to receive code to update pass

Sign Up
collect information of
public users

as a
public user

check the account number against our database to detect the

> Commissioner

profile page

if postal code
is provided

Only show
the province
that they're
responsible for

clicks on one
block to zoom in

(From latest to oldest)

- 2026-Aug-1 3pm

Hand-drawn sketch of a web application interface for "My Submissions". The interface includes a header with the title "My Submissions", a sidebar with a "Home" link, a main content area showing a map of "Toronto Metro" with a highlighted blue area, and a footer with a "Back to map" link.

click one riding

Reference id : 119583

Q. You discover an unreported village.

Answer: you will report it only if it meets the following criteria:

1. > 1000 pop.
2. > 10000 pop.
3. > 10000 pop.
4. > 10000 pop.

My Submissions

Data shows here:

- 1) population
- 2) population distribution
- 3) biogeography
- 4) community of plants

Draw current boundary

Click the video to redraw or make any submissions

Full name	Collection	Comments	Avatar/Link	Tags	Date	Assign
Eric Liu	Yes	I think	1st time		2018-04-01	

click the row

[illegible][illegible]

Click "✓", provided the chosen riding
is not disconnected

My Subsystems

GIS

Data

Output

Analysis

1) population 120000000
2) population 100000000
3) demographics
4) communication of data

GIS

If they cancel the counter-proposal,
we'll still keep the shape of the
drawing so they can make adjustment.